Set Theory for Computer Science

Week 1: Sets

Sets: definition and notation

Fundamental set operations

Venn diagrams and partitions

The algebra of sets

Definition and notation of sets

A **set** is an unordered collection of elements

Examples

$$\begin{aligned} \mathit{DaysOfWeek} &:= \{\mathit{Mon}, \mathit{Tue}, \mathit{Wed}, \mathit{Thu}, \mathit{Fri}, \mathit{Sat}, \mathit{Sun}\} \\ A &:= \{1, 2, 3\} \\ \mathit{Digits} &:= \{0, 1, \ldots, 9\} \end{aligned}$$

$$\begin{aligned} \mathbb{N} &:= \{0, 1, 2, 3, \ldots\} && \text{natural numbers} \\ \mathbb{Z} &:= \{\ldots, -2, -1, 0, 1, 2, \ldots\} && \text{integer numbers} \end{aligned}$$

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Prototype notation

$$\begin{aligned} \mathit{MultiplesOf2} &:= \{2k \colon k \text{ a natural number}\} = \{0, 2, 4, \ldots\} \\ \mathit{Months} &:= \{x \colon x \text{ is a month}\} \end{aligned}$$

Element and subset

Element

$a \in A$	means	a is an element of the set A
$a \notin A$	means	a is not an element of the set A

Subset

$A \subseteq B$	means	all elements of A are elements of B
$A \nsubseteq B$	means	at least one element of A is not in B

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Examples

$4 \in \{1, 2, 3, 4\}$	$\{2, 3\} \subseteq \{1, 2, 3, 4\}$
$5 \notin \{1, 2, 3, 4\}$	$\{2, 5\} \nsubseteq \{1, 2, 3, 4\}$

Set equality and the empty set

Definition of set equality

$A = B \qquad \iff \qquad A \subseteq B$ and $B \subseteq A$

Examples

$\{1, 2, 3, 4\} = \{4, 3, 1, 2\} = \{4, 3, 3, 2, 1, 2\}$

$\{1, 2, 3, 4\} \neq \{2, 3, 4, 5\}$

Set equality and the empty set

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$A = B \qquad \Longleftrightarrow \qquad A \subseteq B$ and $B \subseteq A$

Examples

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$\{1,2,3,4\} \neq \{2,3,4,5\}$

Definition of empty set

$\varnothing$ or $\{\,\}$ is the (unique) set that has no elements

Number of elements

$\#\{4,3,3,2,1,2\} = \#\{1,2,3,4\} = 4$

$\#\varnothing = 0$

Exercise 1

Write down all subsets of $\{1\}$ and of $\{1, 2\}$

Exercise 2

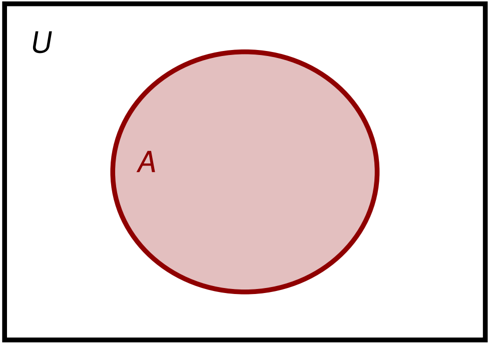
Are the following set inclusions true or false?

(a) $\{1, 3, 5, 7\} \subseteq \{7, 6, 5, 4, 3, 2, 6, 8\}$

(b) $\{1, 3, 5, 7\} \subseteq \{7, 4, 1, 6, 5, 4, 4, 3\}$

Universe

All our sets are subsets of a **universe** U



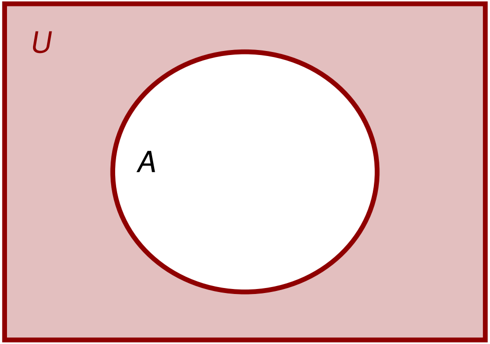
Example

U: all 4-letter words

A: words with a

U defines the *context* we work in to avoid [Russell's Paradox]

Complement of A



Example

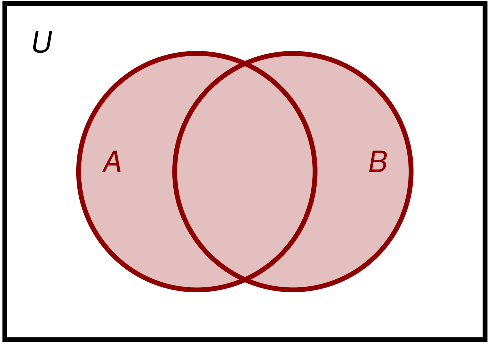
U: all 4-letter words

A: words with a

A':
words with *no* a

$$A' := \{x \in U \colon x \notin A\}$$

Union of A and B



Example

U: all 4-letter words

A: words with $\mathtt{a}$

B: words with $\mathtt{b}$

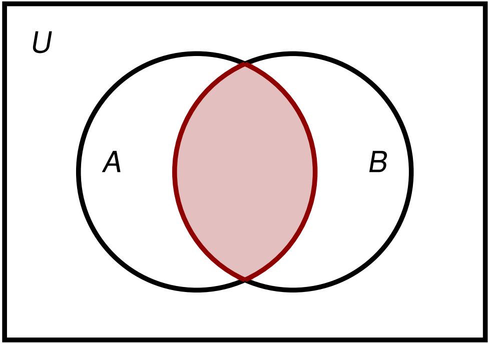
$A \cup B$:
words with $\mathtt{a}$ *or* $\mathtt{b}$

(inclusive *or*)

$$A \cup B := \{x \in U \colon x \in A \text{ or } x \in B\}$$

Union and intersection

Intersection of A and B



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words with $\mathtt{a}$ *and* $\mathtt{b}$

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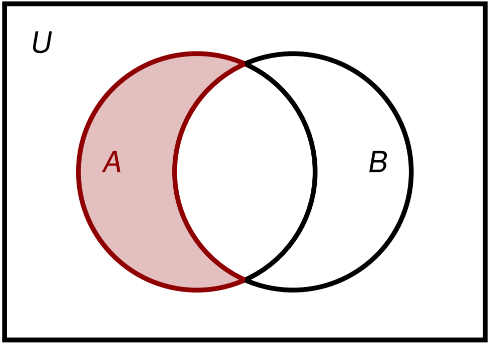
A minus B



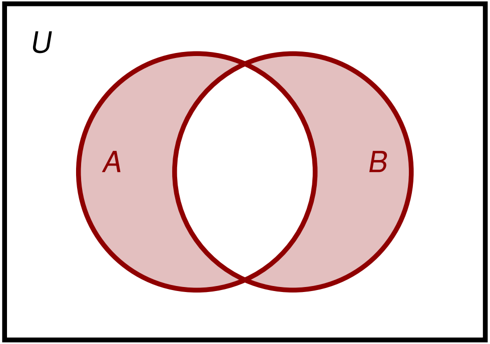
Example

U: all 4-letter words
A: words with $\mathtt{a}$
B: words with $\mathtt{b}$

$A \setminus B$:
words with $\mathtt{a}$ *but no* $\mathtt{b}$

$$A \setminus B := \{x \in A : x \notin B\} = A \cap B'$$

Symmetric difference of A and B



Example

U: all 4-letter words
A: words with $\mathtt{a}$
B: words with $\mathtt{b}$

$A \triangle B$:
words with $\mathtt{a}$ *or* $\mathtt{b}$
but not both
(exclusive *or*)

$$\begin{aligned} A \triangle B &:= (A \cup B) \cap (A \cap B)' \\ &= (A \setminus B) \cup (B \setminus A) \end{aligned}$$

Exercise

We are given the data:

$A := \{0, 1, 2, 3\}$

$B := \{2, 3, 4, 5\}$

The universe is $\mathbb{N}$

Determine the following sets:

$A \cup B =$

$A \cap B =$

$A \setminus B =$

$B \setminus A =$

$A' =$

$B' =$

$A \triangle B =$

break

Sets: definition and notation

Fundamental set operations

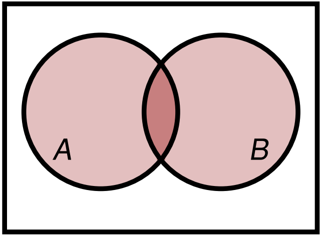
Venn diagrams and partitions

The algebra of sets

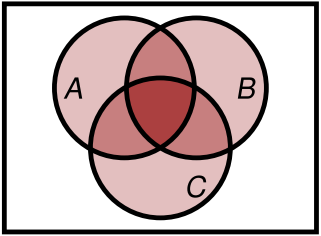
Venn diagrams

Venn diagram: abstract visualisation of relations between sets

2 sets $\Rightarrow$ 4 regions

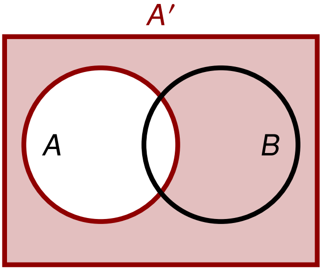


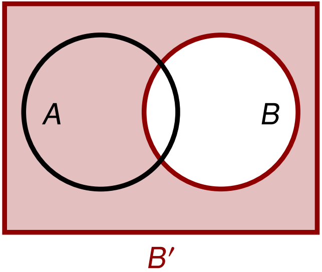
3 sets $\Rightarrow$ 8 regions



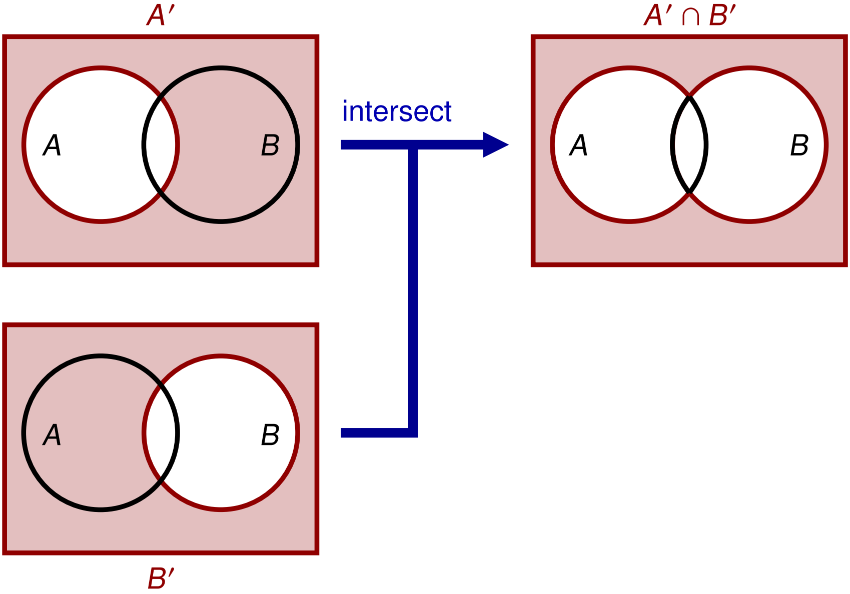
Every region in a Venn diagram can be empty or non-empty

Venn diagrams: De Morgan's law $A' \cap B' = (A \cup B)'$

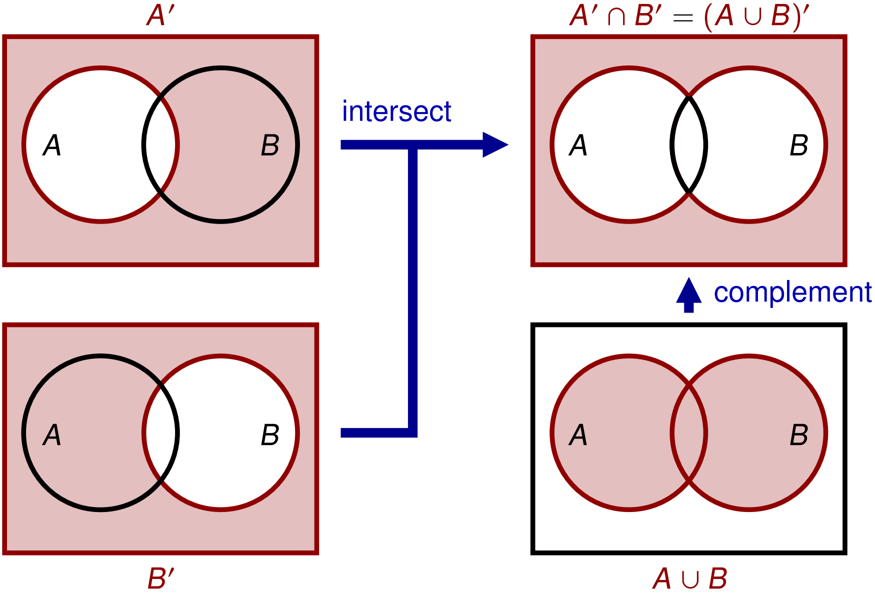




Venn diagrams: De Morgan's law $A' \cap B' = (A \cup B)'$

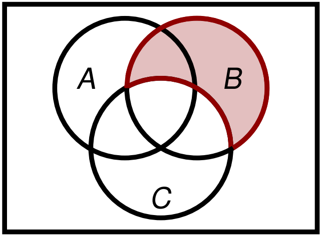


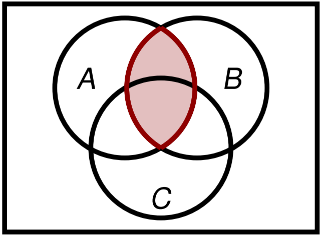
Venn diagrams: De Morgan's law $A' \cap B' = (A \cup B)'$



Venn diagrams: $A \cap (B \setminus C) = (A \cap B) \setminus C$

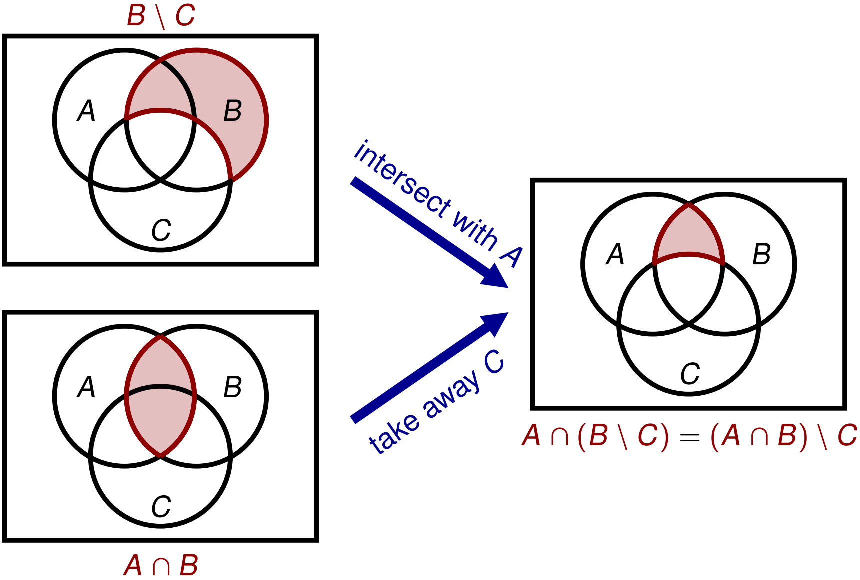
$B \setminus C$





$A \cap B$

Venn diagrams: $A \cap (B \setminus C) = (A \cap B) \setminus C$



Venn diagrams: counting elements

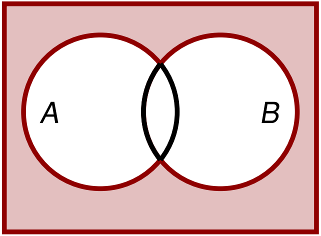
Data

U has 15 elements

$A \cap B$ has 3 elements

A has 10 elements

B has 6 elements



Problem

Determine $\#(A \cup B)'$

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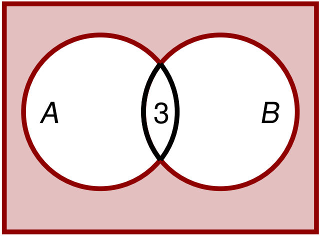
Determine $\#(A \cup B)'$



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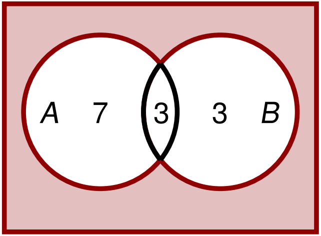
Problem

Determine $\#(A \cup B)'$



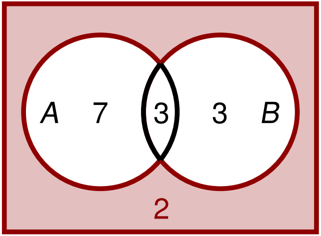
Data

U has 15 elements ✓

$A \cap B$ has 3 elements ✓

A has 10 elements ✓

B has 6 elements ✓



Problem

Determine $\#(A \cup B)'$

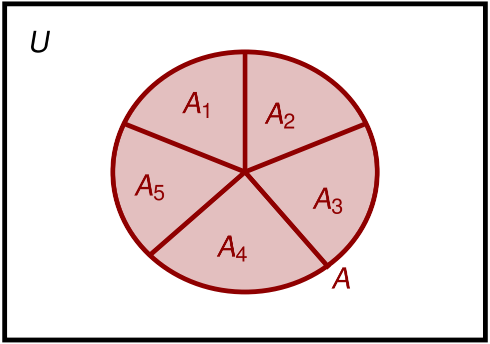
Answer

$\#(A \cup B)' = 2$

Partitions

A **partition** of A is a collection $\{A_1, \ldots, A_n\}$ of **parts** $A_i \subseteq A$ such that

- the parts are *not empty*
- the parts *do not overlap* (are *disjoint*)
- the parts together *cover* A



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Rule of sum for partitions

$\#A = \#A_1 + \#A_2 + \cdots + \#A_n$ "A is the sum of the parts"

Example

$\{0, 1, 2, \ldots, 9\} \quad \Rightarrow$ partition $\{\{2, 4, 6, 8\}, \{0, 1, 9\}, \{3, 5, 7\}\}$

$$10 = 4 + 3 + 3$$

Exercise 1

Verify using Venn diagrams that

$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$

Exercise 2

We are given the data:

$\#U = 28$

$\#A = \#B = \#C = 12$

$\#(A \cap B) = \#(A \cap C) = \#(B \cap C) = 4$

$\#(A \cap B \cap C) = 1$

Use Venn diagrams to determine

$\#(A \cup B \cup C)$

$\#(B \cup C)'$

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The algebra of sets

Commutativity

$A \cap B = B \cap A$

$A \cup B = B \cup A$

Associativity

$A \cap (B \cap C) = (A \cap B) \cap C$

$A \cup (B \cup C) = (A \cup B) \cup C$

Complement

$A \cap A' = \varnothing$

$A \cup A' = U$

Identity

$A \cap U = A$

$A \cup \varnothing = A$

Domination

$A \cap \varnothing = \varnothing$

$A \cup U = U$

Involution

$(A')' = A$

Idempotence

$A \cap A = A$

$A \cup A = A$

Definitions

$A \setminus B = A \cap B'$

$A \triangle B = (A \cup B) \cap (A \cap B)'$

De Morgan

$(A \cap B)' = A' \cup B'$

$(A \cup B)' = A' \cap B'$

Distributivity

$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$

$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$

Repeated unions and intersections

Commutativity

$A \cap B = B \cap A$

$A \cup B = B \cup A$

Associativity

$A \cap (B \cap C) = (A \cap B) \cap C$

$A \cup (B \cup C) = (A \cup B) \cup C$

Reorder rule

In a repeated union or repeated intersection
you can omit brackets and change the order

Examples

- $A \cap (B \cap C) = A \cap B \cap C$
- $A \cap (B \cap C) = (B \cap A) \cap C$
- $(A \cup B) \cup (C \cup D) = D \cup C \cup B \cup A$
- $(A \cup B) \cup (C \cup D) = (C \cup A) \cup (D \cup B)$

Checking set equality using the algebra of sets

Example

Check that $(A' \cap B)' \cup (C \cap A')' = A \cup (B \cap C)'$

Checking set equality using the algebra of sets

Example

Check that $(A' \cap B)' \cup (C \cap A')' = A \cup (B \cap C)'$

Solution

$$
\begin{aligned}
&(A' \cap B)' \cup (C \cap A')' \\
&\qquad = ((A')' \cup B') \cup (C' \cup (A')') && \text{(De Morgan)} \\
&\qquad = (A \cup B') \cup (C' \cup A) && \text{(involution)} \\
&\qquad = (A \cup A) \cup (B' \cup C') && \text{(reorder)} \\
&\qquad = A \cup (B' \cup C') && \text{(idempotence)} \\
&\qquad = A \cup (B \cap C)' && \text{(De Morgan)}
\end{aligned}
$$

Exercise

Check that $(A' \cap (A \cup B'))' = A \cup B$ using the algebra of sets